

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

Ans: -

No.	BIOS/POST Error Message	Hardware / Configuration Issue
1	CMOS Checksum Error	Weak CMOS battery or incorrect BIOS settings
2	Keyboard Error	Keyboard or USB connection problem
3	Memory Error / RAM Error	Loose, faulty, or incompatible RAM
4	CPU Fan Error	CPU fan connection or cooling problem
5	No Boot Device Found	HDD/SSD, boot order, or OS problem

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Ans:-

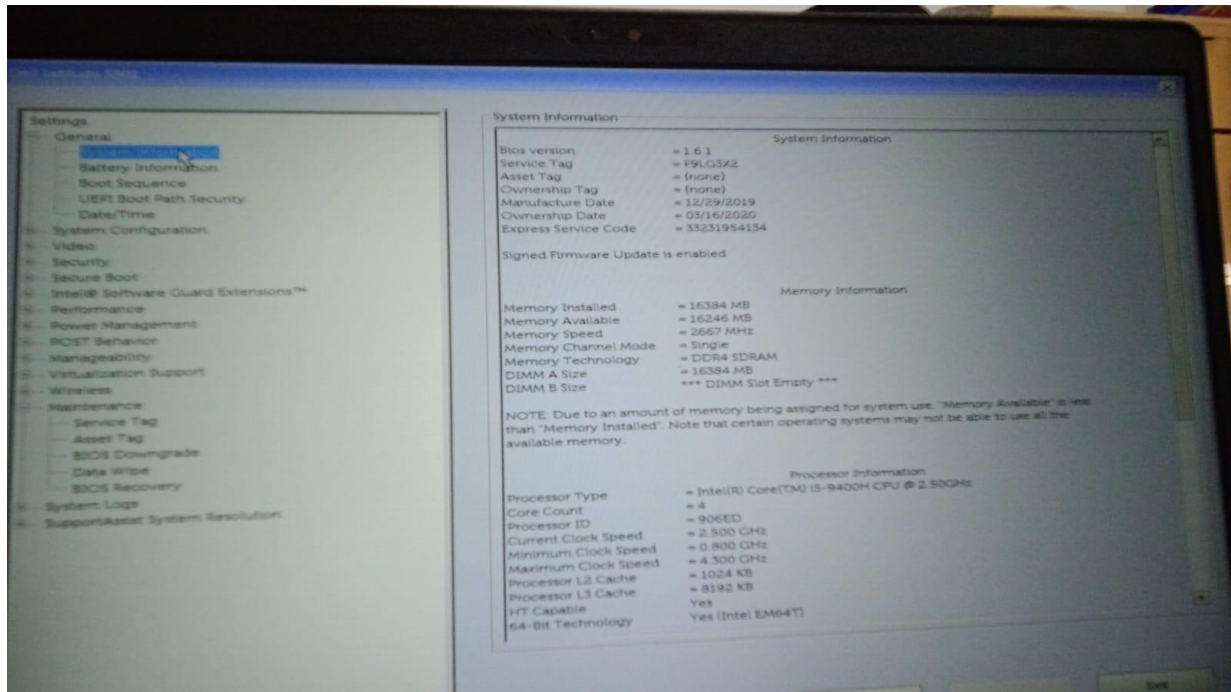
AMI BIOS – 3 Beep

Meaning: 3 beeps usually indicate a base memory (RAM) failure, specifically a failure in the first 64 KB of memory.

1. **Cause:** The RAM may be loose, faulty, incompatible, or have poor contact.
2. **Troubleshooting Step 1:** Turn off and unplug the PC, then remove and firmly reseat the RAM.
3. **Troubleshooting Step 2:** Test one RAM stick at a time and try a different RAM slot or a known-good compatible RAM module.
4. **Note:** Beep codes can vary by motherboard/BIOS version, so check the specific motherboard manual if possible.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

Ans: -



4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

Ans: -

Meaning:

- The BIOS detected that the saved CMOS/BIOS settings are corrupted or incorrect.
- The system has loaded the default BIOS settings.
- A weak or dead CMOS battery is a common cause.

How to Resolve:

- Turn off the PC and unplug the power cable.
- Open the computer case.
- Locate the CMOS battery (usually a CR2032 coin-cell battery).
- Remove the old battery and replace it with a new one.
- Start the PC and enter BIOS Setup.
- Set the correct date, time, and required boot settings.
- Save the BIOS settings and restart the computer.
- If the error continues, reset/clear CMOS and check for motherboard or BIOS problems.